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A multi-string and wireline-coring drilling robot for planetary regolith sampling

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Abstract

Planetary regolith sampling is essential to deep space exploration missions, motivating researchers to develop innovative sampling techniques for efficient regolith sample acquisition. The Multi-string and Wireline-coring Drilling Robot (MWDR) for planetary regolith sampling is a bottom-hole continuous sampling technology that can efficiently and safely perform deep drilling on extraterrestrial bodies and rapidly obtain bottom samples through a wireline, eliminating the need for astronaut operations. This paper presents the design considerations and development process of an MWDR planetary regolith sampling, from conceptual design to prototype testing, aimed at achieving automatic deep regolith sampling. First, the paper outlines the essential functions of the MWDR based on the requirements for automatic regolith sampling and describes the components that fulfill these functions. Next, the mechanisms and control system of a novel MWDR are proposed to perform automatic regolith sampling tasks, including drilling components for penetration, rotation, and clamping motions, a manipulation component for grabbing and transferring multiple drill tools, and a coring component for collecting regolith samples. A feasible drilling and sampling strategy is then developed, with detailed motion planning for each mechanism to coordinate their movements. Finally, an experimental prototype is developed to conduct automatic drilling tests, validating the functionality of the proposed mechanism system for the MWDR.
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Keywords: Planetary regolith sampling; Multi-string drilling robot; Wireline coring

1. Introduction

Deep space exploration missions represent significant milestones in the history of human space science and serve as ambitious objectives driving future technological advancements. Through deep space exploration, humanity can enhance its understanding of the origins and evolution of the solar system, while also uncovering the mysteries surrounding the origin and evolution of life. These insights offer essential scientific foundations for the sustainable development of both Earth and human society. Moreover, deep space exploration holds tremendous potential for utilizing space environments, energy, and resources, thereby opening new pathways to address Earth's resource shortages and advancing human society toward a space-faring civilization (Zhang et al., 2021a; Zhang et al., 2023a). Against this backdrop, extraterrestrial bodies have become some of the most crucial targets for deep space exploration missions. In recent years, the exploration of extraterrestrial bodies, including the Moon and Mars, has garnered

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substantial attention within the global aerospace community (Smith et al., 2020; Sibeck et al., 2014; Li et al., 2021). These celestial bodies are frequently covered by a planetary regolith layer, with thicknesses ranging from several meters to tens of meters (Spray, 2016). Planetary regolith, formed through prolonged processes such as micrometeoroid impacts, solar wind irradiation, and cosmic ray exposure, serves as a crucial record of the surface evolution of extraterrestrial bodies. Acting as a transitional zone between the lithosphere and outer space, planetary regolith preserves critical information about the composition and distribution of surface lithospheres, while also providing direct evidence for studies of solar system material cycles, space radiation effects, and solar wind composition (Badescu and Zacny, 2018; Team, 1969). Therefore, sampling and analyzing planetary regolith is a key method for investigating extraterrestrial bodies. These endeavors not only reveal the formation and evolutionary history of planetary bodies but also validate various deep space exploration theories, thus laying the foundation for more complex future missions. Through mineralogical and chemical analyses of regolith samples, planetary volcanic activities and impact histories can be inferred with precision, and even processes from the early evolution of the solar system can be uncovered. Additionally, regolith sampling drives advancements in the technological aspects of deep space missions, including sampling technologies, encapsulation and storage techniques, and extraterrestrial sample return capabilities. Advancements in these technologies will significantly propel future deep space exploration programs. Regolith sampling plays a pivotal role in deep space exploration missions and serves as a crucial link in advancing human understanding of extraterrestrial bodies.

However, the working environment on extraterrestrial bodies is extremely harsh, characterized by low gravity, large temperature variations, high vacuum, intense radiation, and complex geology (Zhang et al., 2022a; Zacny and Cooper, 2006). These conditions introduce substantial uncertainties and challenges for astronauts working on the surface, making sampling activities significantly more difficult. A viable solution is to deploy robots to perform sampling tasks in place of astronauts, which can greatly enhance both sampling efficiency and cost-effectiveness. This has been demonstrated in numerous robotic sampling missions (Paulsen et al., 2006; Zhang et al., 2021b). For future lunar exploration missions, robotic sampling is an inevitable trend and long-term goal for utilizing lunar in situ resources to establish lunar bases (Zacny, 2012; Zhou et al., 2019). In 2025, the Artemis program proposed by NASA is expected to be the most significant international manned lunar exploration mission. As part of this program, Honeybee Robotics has developed the latest lunar regolith sampling drill, Trident, designed to collect lunar regolith samples from beneath the surface of the lunar south pole at depths of up to 1 m, where water ice is likely to be abundant (Zacny et al., 2024; Zacny et al., 2022; Chu et al., 2024). The drill employs a single drill pipe to penetrate the regolith, capturing the sample within the spiral grooves. The sample is then extracted and cleaned to complete the sampling process. In the near future, Roscosmos plans to launch the Luna-27 mission between 2027 and 2028, which will also carry the Proseed drilling robot, capable of drilling to a depth of 1.2 m (Heather et al., 2021; Trautner et al., 2024; Heather et al., 2022). The mission aims to assess the content and composition of volatile substances in the regolith beneath the lunar south pole surface. The key difference is that this drilling system collects regolith samples through a cavity within the drill bit, enabling the drill bit to maintain axial drilling motion relative to the drill pipe. Based on the latest advancements in robotic sampling research, space agencies worldwide are focused on exploring previously unexplored regions of the Moon and Mars. While deep drilling is a key goal for every sampling mission, achieving successful shallow drilling has already marked a significant advancement, driven by curiosity about unexplored regions and unpredictable geological conditions. In recent years, countries worldwide have developed increasingly ambitious space plans, aiming to establish lunar research stations on the Moon for long-term, comprehensive in situ scientific exploration. Therefore, further pioneering innovations and technological validation of robotic deep drilling in this context represent a highly promising field of scientific research.

In response to the increasing demands of scientific research tasks and commercial exploration activities, significant progress has been made in the development of extraterrestrial robotic sampling. Various sampling mechanisms have been developed for conducting regolith sampling in diverse scenarios, including drilling, excavating, projecting, ultrasonic, pneumatic, grabbing, abrasive, and bio-inspired sampling robots (Zhang et al., 2019; Wei et al., 2021). Among these, drilling robots are particularly suited for collecting samples at greater depths, effectively preserving the vertical stratification of the samples (Bar-Cohen and Zacny, 2009; Wang et al., 2024). This approach is especially favored by researchers, as deep regolith, shielded by the upper layers, retains more valuable in situ information about the celestial body. To enable sampling of deeper regolith layers, two robotic deep drilling methods have been proposed: submersible drilling and multi-string drilling (Zhang et al., 2022b). Submersible drilling robots exhibit high autonomous mobility and lower reaction forces due to their compact size. Mounted on landers or rovers as sub-robots, they can autonomously drill from the surface to deeper layers and collect regolith samples using their advanced mobility. In comparison, multi-string drilling robots offer superior penetration capabilities, enabling them to break through rock layers and achieve greater drilling depths. For the same drilling depth, they typically collect more regolith samples while preserving the vertical stratification of the samples. Therefore, multi-string drilling robots hold considerable application value in planetary sampling missions.

Currently, a variety of multi-string drilling robots for planetary regolith sampling have been validated on the ground. Ding et al. (Ding et al., 2009) developed a multi-string deep sampler for lunar regolith collection, consisting primarily of a pipe-changing mechanism, a drilling mechanism, and drill pipes. This single-pipe automatic sampler performs deep drilling by sequentially connecting drill pipes fixed to the pipe-changing mechanism through the coordinated motion of both mechanisms, offering excellent scalability. Hou et al. (Hou et al., 2019; Liang et al., 2016) first proposed multi-string drilling technology with airflow-assisted sampling and developed a multi-string pneumatic sampling device for drilling to a depth of 5 m on Mars. The device primarily consists of a feeding mechanism, a rotating mechanism, a clamping mechanism, and an indexing mechanism. Multiple double-layer hollow drill pipes are stored on the indexing mechanism and can be automatically assembled in coordination with the clamping mechanism. An airflow passage is established by installing an airflow coupling and a sample collection device at the top of the drill pipe. High-pressure airflow is injected into the annular gap of the drill pipe, guiding Martian regolith to rise from the bottom along the wall surface and enter the sample collection device. Zhang et al. (Zhang et al., 2023b) developed an automatic-assisted drill system for deep extraterrestrial regolith sampling. This system can sample the regolith layer using sectional detachable core pipes, drill pipes, and casings, and operates in both manually assisted and unmanned automatic modes. However, it employs a traditional coring method that requires the core pipe to be lifted each time to retrieve samples. NASA funded Honeybee Robotics to develop a 10-m-class Mars robotic drilling system. This system is a highly automatic, low-power deep drilling and coring platform (Cannon et al., 2007; Stoker et al., 2008). To enhance drilling efficiency under continuous coring conditions, the system retrieves a core sample every 25 cm from the borehole, simultaneously bringing the drill cuttings to the surface. This method reduces redundant drilling pressure and torque generated by the cuttings. ESA funded Italian company Galileo Avionica to develop the DeeDri drilling system, one of the latest drilling robots, which is mounted on the ExoMars rover for 2-m deep regolith sampling (Altieri et al., 2023; Re et al., 2008; Van Winnendael et al., 2005). Unlike typical multi-string sampling systems, this drilling system integrates several sensors within the drill pipe to monitor the borehole wall and drilling conditions. Consequently, the sampling space is highly constrained, and the system is designed to obtain only partial samples from the target depth.

Recent research advancements indicate that multi-string drilling robots are typically equipped with dual-pipe or single-pipe drill tools for deep sampling. Two primary coring methods are employed during core retrieval: lifting coring and wireline coring. These two coring methods differ significantly in their operation. In lifting coring, once each drill pipe has fully penetrated the planetary surface and filled the core pipe with samples, retrieving the bottom core pipe and storing the core samples necessitate the extraction and disassembly of all drill pipes from the borehole. Subsequently, the drill pipes must be reassembled and extended to the previous depth before a new pipe can be added to continue drilling. While this procedure is feasible for single-pass and shallow coring, the frequent disassembly and reassembly of drill tools become increasingly intricate and time-consuming for deep planetary-regolith coring operations. Conversely, wireline coring eliminates the need to lift the entire drill string to retrieve bottom core samples. Instead, once a drill pipe penetrates, the core pipe containing samples is directly retrieved from the borehole bottom using an overshot. Following this, a new core pipe is installed, and a new drill pipe is connected to allow drilling operations to proceed until the target depth is reached (Liu et al., 2024). Wireline coring requires lifting the entire drill string for core retrieval only under certain circumstances, such as drill bit wear or borehole blockage. This feature makes it particularly advantageous for planetary regolith sampling, as it reduces the time required for tool installation in unmanned operations in complex planetary environments, thus mitigating potential safety hazards associated with redundant operations (Ashena et al., 2016; Ali et al., 2014). Moreover, the reduced need for drill string hoisting effectively safeguards the drill bit and borehole walls while minimizing disturbances to core samples. However, operational complexity alone does not fully account for the time advantages of a coring method, as increasing the speed of certain operations may reduce time differences. Nevertheless, this does not fundamentally resolve the issue. This is because vibrations and safety concerns caused by high-speed movements make raising the entire drill string a heavy-load issue for the mechanical system, whereas wireline coring typically allows for higher lifting speeds (Liu et al., 2023; Ye et al., 2024).

Wireline coring based on dual-pipe drilling and bottom-hole sampling is a preferred method for more safely and efficiently completing deep drilling activities, as it enables the rapid acquisition of bottom core samples while minimizing the extra energy consumption and uncertainties associated with repeatedly lifting the drill pipe (Ashena and Thonhauser, 2018; Kieft et al., 2007). Although multi-string drilling robots have been extensively researched and validated on Earth, none of them are capable of performing continuous core extraction through the use of wireline in the drill string. This would enable the robot to obtain regolith samples generated during drilling without raising the drill string, thus ensuring the continuity of samples in vertical stratigraphy. Therefore, wireline coring, as a key innovative feature, has been creatively integrated into our design of the robotic system, leading to the development of a novel multi-string and wireline coring drilling robot (MWDR) for planetary regolith sampling. The configuration and layout of the MWDR mechanisms determine its sampling process and sequence, with variations in these processes and sequences directly

impacting the complexity of movement and the continuity of operations. To achieve fully automatic sampling of planetary surfaces and eliminate the need for manual astronaut intervention, this study systematically analyzes the design requirements for a fully automatic MWDR for planetary regolith sampling. A structured approach to the design of the mechanism system is proposed. The study reveals the coordinated motion between the different mechanisms in the MWDR during the planetary regolith sampling process. These coordinated movements are crucial for achieving a more complete and compact fully automatic sampling process, and the motion capabilities of the robot have been preliminarily validated.

The subsequent sections of this paper are organized as follows: Section II presents the design considerations for the MWDR, focusing on functional requirements and the composition of mechanical components. Section III provides a detailed description of the design aspects of the mechanism components and control system of the MWDR. Section IV delves into a feasible motion strategy, proposing multiple motion sequences to coordinate the movement of mechanisms. Section V introduces the prototype development process and presents the functional verification results. Section VI provides a concluding summary of the study.

2. Design considerations

The core consideration in the design process is to ensure that the MWDR possesses high flexibility and adaptability to autonomously perform sampling tasks in extreme and unpredictable extraterrestrial environments. At the same time, it is essential to ensure that the system components are as streamlined as possible, providing a reliable technical foundation for subsequent optimization in terms of size, weight, and performance. Therefore, it is necessary to assign multiple functions to each component and then proceed with a modular design approach. This section summarizes the essential functional requirements of the MWDR and proposes component composition solutions based on these requirements.

2.1. Functional requirements

Regolith samples from different depths exhibit significant variations in mineral composition, chemical content, particle morphology, distribution patterns, and even moisture content (McKay et al., 1991; Hui et al., 2013). To fully investigate the important scientific information contained in regolith at various depths, it is essential to continuously acquire samples from the surface to deeper layers, preserving as much stratigraphic information as possible. Unmanned deep drilling presents numerous technical challenges, with the primary goal being to ensure that a drilling robot can continuously collect, retrieve, and store regolith samples without requiring astronaut intervention (Tang et al., 2018; Tang et al., 2017). Based on the principles of coring operations, the MWDR for planetary regolith sampling must be equipped with both drill pipes and core pipes. The main body should be easily deployable on the lander. It penetrates the strata through the rotational force of the drill pipe and drill bit, continuously adding drill pipes. It acquires core samples by timely raising and lowering the core pipe in segments while continuously replacing it. After sampling is completed, the entire drilling apparatus must be extracted and stored on the lander. Therefore, multiple mechanisms must operate at designated locations to simulate manual drilling. The main functions of the MWDR include two modules: drill pipe operation and core pipe operation. Specifically, it must meet the following functional requirements (Fig. 1):

(1) *Penetration and rotation:* Penetration and rotation are the fundamental functions of an MWDR for regolith sampling (Anttila, 2005). It must have the capability to penetrate the surface strata of celestial bodies by driving drill pipes into the regolith layer and extracting them from the drilled borehole. Simultaneously, it must be equipped with rotation motion to drive the drill bit, cutting through the regolith layer and facilitating the coring process in conjunction with the penetration motion.
(2) *Drill pipe manipulation:* For shallow penetration, a single drill pipe is adequate to reach the desired sampling depth (Zacny et al., 2013). However, for deeper penetration up to tens or even hundreds of meters, it is necessary to connect a long drill string composed of multiple pipes. This necessitates that the MWDR perform automatic manipulation of multiple drill pipes on the planetary surface, including grabbing, clamping, (dis) connection, transfer, and storage.
(3) *Core pipe manipulation:* Hollow core pipes are the essential components for retaining sampled regolith for scientific or engineering purposes. To conduct deep regolith sampling, the core pipe must be divided into multiple segments because it cannot traverse long distances within a narrow borehole. The MWDR must autonomously manipulate the core pipe on the planetary surface, including tasks such as grabbing, clamping, positioning, retrieval, and storage.

2.2. Component composition

In traditional multi-sting samplers, retrieving core samples requires the complete extraction of all drill pipes. This is because the core samples cannot pass through the hollow drill string (Zacny et al., 2008; Knez and Khalilidermani, 2021; Allton, 1989). In the complex surface environment of celestial bodies, frequently lifting drill tools is highly inefficient in terms of energy use and management, and it can lead to borehole collapse. While installing a casing is an effective method for wall protection, utilizing drill pipes

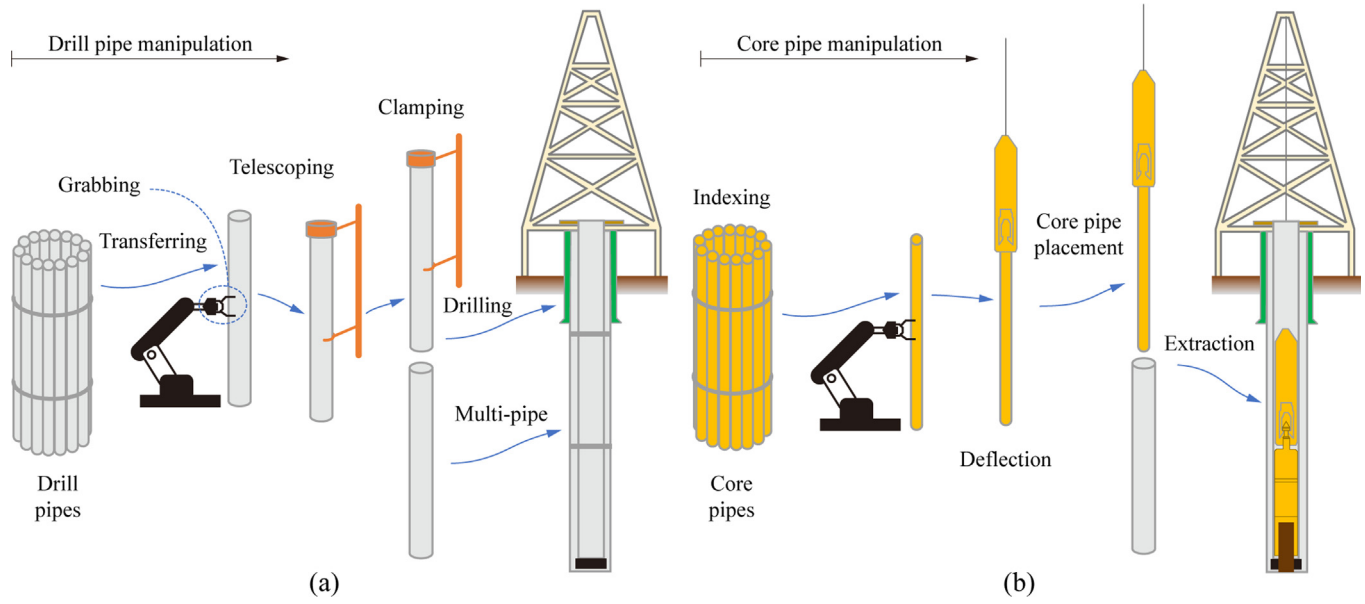


Fig. 1. System function of an MWDR. (a) Drill pipe manipulation. (b) Core pipe manipulation.

as casings can save the weight of carrying additional casings (Sawaryn et al., 2018; Zacny et al., 2016). To optimize sampling efficiency, a more effective strategy is the wireline coring method, which involves sampling regolith using core pipes and withdrawing only the drilled core (inner) pipes while leaving the drill (outer) pipes in place. Based on the aforementioned functional requirements, the entire mechanism system can be divided into three components: (i) a drilling component for penetration, rotation, and clamping; (ii) a manipulation component for grabbing, transferring, and storing drill/core pipes; (iii) a coring component for deflecting and extracting core pipes. The details of the DOFs for these components are listed in Table 1.

3. Robotic system design

The system design of the MWDR primarily consists of two main parts: the mechanism component and the control system. The mechanism component design divides the system into three key submodules: the drilling component, the manipulation component, and the coring component. The design of each component centers around its core function, with a focus on the key structures and working principles. Additionally, the hardware composition of the control system is elaborated in detail. Each carefully designed component can be easily integrated into a testing framework, thus forming a powerful and efficient robotic system.

3.1. Drilling component

The drilling assembly is an automatic platform with only drilling functions. It primarily consists of a penetration unit, a rotation unit, and two clamping units. The penetration unit vertically translates the rotation and top clamping units into or out of the regolith. The rotation unit is responsible for rotating the drill pipes to cut the regolith. The top clamping unit operates in conjunction with the manipulation component to mount the drill pipe, enabling it to clamp the drill pipe during drilling and withdrawal, and coordinates with the bottom clamping unit to connect and disconnect the drill string.

Table 1
Description of each DOF for the MWDR.

Component	DOF	Symbol	Function details
Drilling	Penetration	*p*	Penetrating and retracting drill pipes during drilling
	Rotation	*r*	Rotating drill pipes for regolith cutting and pipe (dis) connection
	Clamping top	*ct*	Clamping drill/core pipes during drilling and pipe withdrawal
	Clamping bottom	*cb*	Clamping the lower drill pipe for (dis) connection
Manipulation	Grabbing	*g*	Grabbing drill pipes for clamping, transferring, and storing
	Telescoping	*te*	Extending a gripper during drill/core pipe manipulation
	Transferring	*tr*	Rotating a gripper during drill/core pipe manipulation
	Indexing	*i*	Indexing a magazine for drill/core pipe selection and storage
Coring	Deflection	*d*	Extending an overshot in and out during core pipe withdrawal
	Extraction	*e*	Extracting an overshot to collect core pipes by wireline

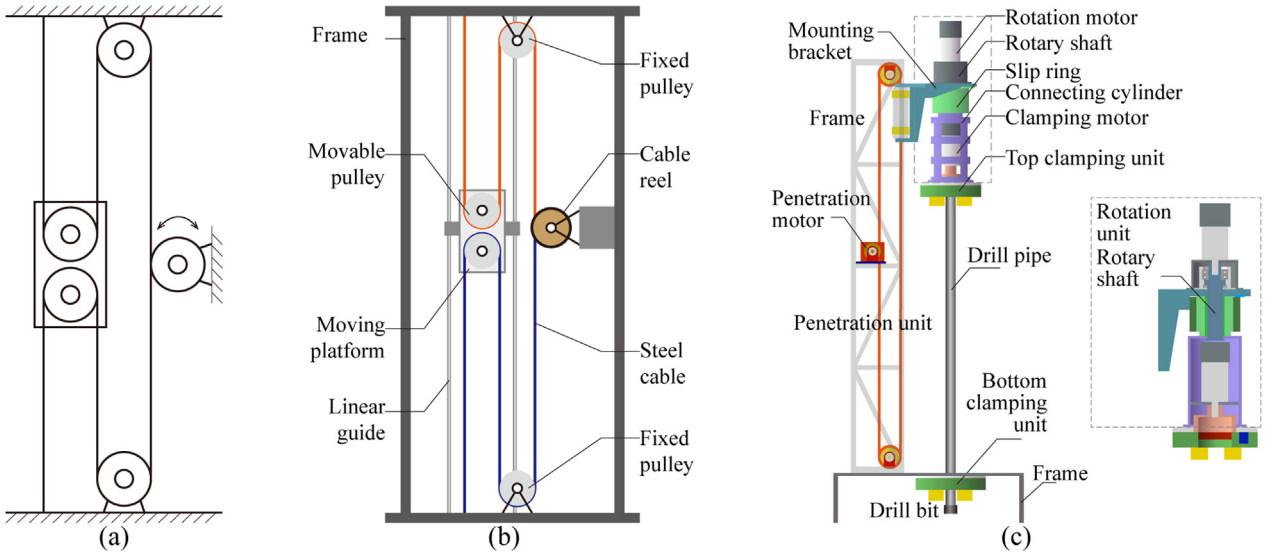


Fig. 2. Penetration and rotation units design. (a) Mechanism configuration. (b) Mechanism principle. (c) 3D model.

3.1.1. Penetration and rotation units

The penetration unit (Fig. 2) employs a cable-driven method, providing several advantages, including lightweight design, space efficiency, and long stroke capabilities. To accommodate drilling pressure requirements at various depths, the wireline can be easily replaced with different sizes without requiring adjustments to other components, thus preserving the maximum load capacity of the system. This substantially enhances the flexibility and adaptability of the system. It primarily consists of a penetration motor, a cable reel, two steel cables, two fixed pulleys, two movable pulleys, a moving platform, and two linear guides. A pair of movable pulleys are mounted on the moving platform to amplify the drag force transmitted by the steel cable. Two fixed pulleys are mounted on the frame to alter the direction of the cables. The cable reel is driven by the penetration motor to move the platform, which is guided by two linear guides. When the cable reel rotates in a predetermined direction, one steel cable is wound while the other is released. When the cable reel rotates in the reverse direction, the moving platform moves in the opposite direction.

The rotation unit primarily consists of a rotation motor, a mounting bracket, a slip ring, a rotary shaft, and a connecting cylinder. It can rotate the drill pipes unidirectionally during penetration and in reverse during drill pipe disconnection after sampling. The output shaft of the rotation motor is connected to the top clamping unit. Double-row angular contact ball bearings and thrust ball bearings support the weight of the top clamping unit and absorb the radial vibrations from the drill pipes during penetration. A slip ring is employed to prevent wire winding during rotation.

The concurrent activation of penetration and rotation motions is commonly observed during drilling, as it ensures the continuous application of shear and cutting forces to the bit, facilitating the fragmentation of the regolith. Before initiating the drilling process, the drilling component must be securely clamped to the drill pipe to ensure consistent transmission of the cutting torque. Once the desired depth is reached, the drilling motions are halted. Subsequently, the top clamping unit releases the drill pipe, enabling the drilling component to move upward to its initial position. However, when loosening the threaded connection between drill pipes, it rotates in the opposite direction, and the penetration motion is coordinated with the thread pitch. This coordinated motion ensures that the connection between the drill pipes is loosened efficiently, without causing any binding or jamming.

3.1.2. Clamping unit

The clamping units (Fig. 3) form a novel parallel mechanism, consisting of a top clamping unit and a bottom clamping unit, both having similar structures. Traditional methods for installing drill pipes typically employ threaded connections. While this is feasible, as the drill pipe advances downward, the connection between the drill pipe and the connector becomes progressively tighter, significantly hindering the rapid disassembly of the drill pipe. To address this issue, the top clamping unit is designed to grip the drill pipe and drive its rotary drilling motion. Furthermore, to facilitate the retrieval of the drill string after the sampling task is completed, the bottom clamping unit must be employed. This is necessary because the disassembly of the two drill pipes must take place while the bottom drill pipe remains securely clamped. It primarily consists of a clamping motor, a base frame, a hollow gear, three jaws, and three geared sliders. The jaws and sliders are evenly distributed around the hollow gear, and the clamping movement resembles that of a three-jaw chuck mechanism. The jaws mounted on the sliders are rotated by the hollow gear to clamp or release the drill pipes.

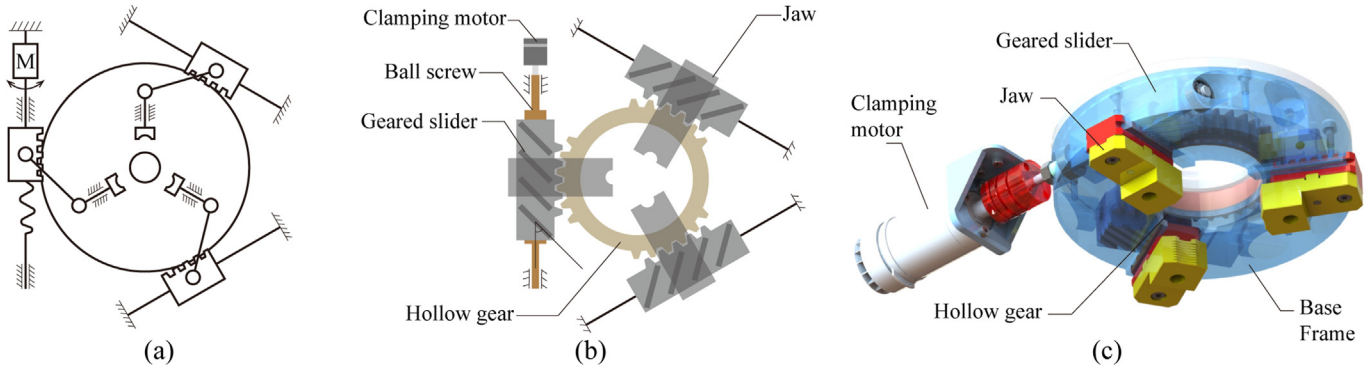


Fig. 3. Clamping unit design. (a) Mechanism configuration. Release state (up). Clamped state (down). (b) Mechanism principle. (c) 3D model.

For the bottom clamping unit, one of the geared sliders is driven by a ball screw connected to the output shaft of the clamping motor. The contact surfaces of the sliders and jaws are designed with grip marks that have an inclination angle, referred to as the transmission angle. The grip marks are aligned, similar to the mating of threads. If they become separated, the jaws can be forced to move by the force exerted by the sliders on the jaws. Therefore, to ensure the identical movement of the three jaws, it is essential to keep the three sliders on the same side of the mounting slots, including the jaws. The hollow gear drives the three sliders to traverse along the circumferential tangent direction, pushing the jaws to move radially via the teeth between the slider and jaw, thus achieving the clamping or releasing movements. The cylindrical base frame supports the components of the overall mechanism and constrains the sliders and jaws. Notably, for the top clamping unit, the hollow gear is directly connected to the output shaft of the clamping motor, making it compact with the rotation unit. The outer surface of the spiral drill pipe is rough, and the top clamping unit is responsible for transmitting the cutting torque. To ensure a stable clamping surface for the top clamping unit, an additional hollow sleeve must be installed at its center. This hollow sleeve is designed to enclose the spiral drill pipe, while being tightly clamped by the clamping unit, thereby ensuring secure and stable clamping during drilling.

3.2. Manipulation component

The length of the drill pipe is the most important factor in determining the sampling depth of MWDRs. Typically, drill pipes are required to penetrate deep into the regolith layer, reaching depths of tens or even hundreds of meters. However, due to constraints related to manufacturing and transportation, producing and employing ultra-long drill pipes in actual operations is not feasible. Currently, drill pipes are typically divided into multiple sections. When a drill pipe becomes fully buried, a new drill pipe must be connected at its top to maintain continuous drilling operations. The number of drill pipes required is determined by the desired drilling depth. A rotary magazine is designed to store multiple drill pipes and core pipes, while a manipulator facilitates the transportation of these pipes between the rotary magazine and the drilling component.

3.2.1. Rotary magazine

The rotary magazine (Fig. 4) mainly consists of an indexing motor, a shaft, two supports, a carousel, and multiple drill and core pipes. All unused pipes are securely stored in the rotary magazine. The carousel is positioned at the bottom of the rotary magazine to prevent pipes from sliding downwards along the axial direction. The carousel is capable of rotating along the intermediate shaft, which is located at the top of the indexing motor. Each time a pipe is drawn or inserted, the carousel rotates to the next target position. Drill and core pipes are alternately arranged around the circular carousel to maximize the utilization of space. Each pipe is secured to the carousel by two clamping jaws arranged in parallel, which are mounted on two supports. The inner surface of each clamping jaw has a groove that matches the outer diameter of the drill or core pipe. The clamping jaws generate elastic deformation to secure the pipes during the drawing or insertion process. If the pipe does not enter the groove, the clamping jaws are forced open, resulting in greater elastic deformation. When the drill pipe enters the groove, the opening angle of the jaws decreases, restricting the radial movement of the drill pipe.

3.2.2. Drill tool manipulator

To transport drill tools between the rotary magazine and the drilling component, the operations of grabbing, drawing, inserting, and rotating pipes must be completed. Quick placement of drill tools between the spindle and their storage positions is a multi-step process that can significantly reduce auxiliary time. Therefore, while ensuring the flexibility of the manipulator, its range of motion is minimized as much as possible to allow each operation to be performed in a stable and controlled manner. To achieve this, a drill tool manipulator (Fig. 5), consisting of a transferring unit, a telescoping unit, and a grabbing unit, is proposed.

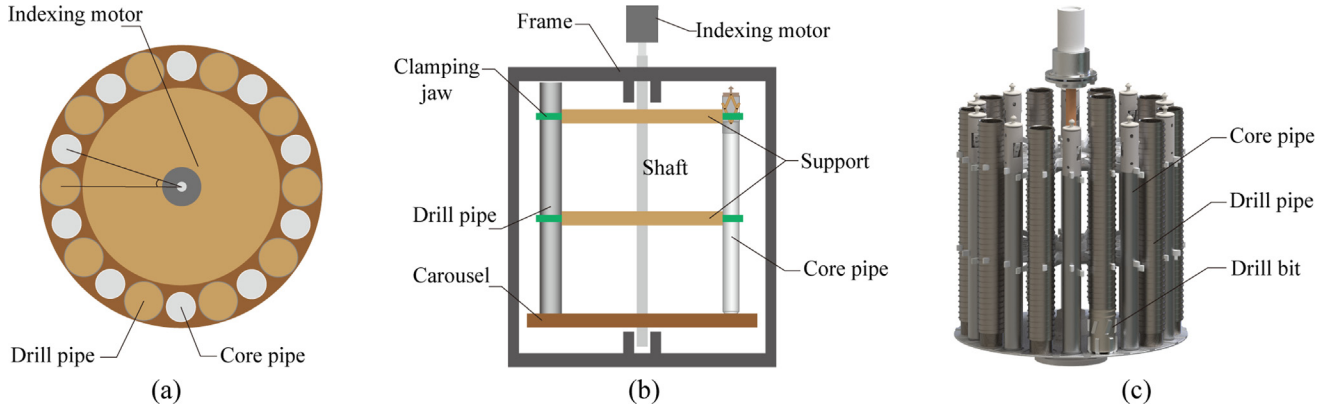


Fig. 4. Rotary magazine design. (a) Pipe arrangement. (b) Mechanism principle. (c) 3D model.

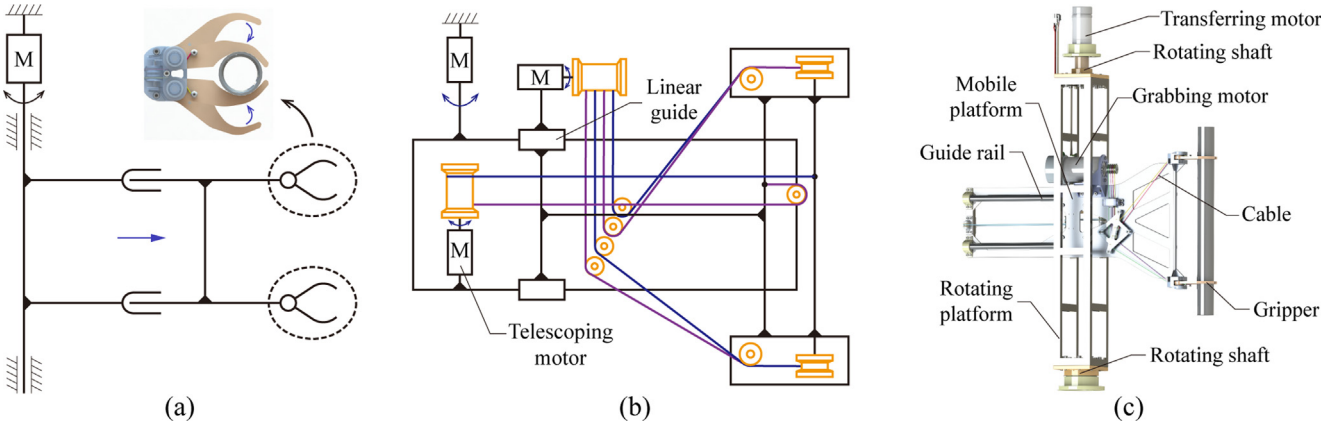


Fig. 5. Drill tool manipulator design. (a) Mechanism configuration. (b) Mechanism principle. (c) 3D model.

The transferring unit primarily consists of a transferring motor, two rotating shafts, and a rotating platform. It is used to rotate and reorient the telescoping unit between the rotary magazine and the drilling component. The rotating platform is connected to the frame by two rotating shafts. The transfer motor is mounted above the platform, with its output shaft connected to the upper shaft to enable the rotation of the manipulator. The telescoping unit, used to draw and insert pipes mounted on the rotary magazine, primarily consists of a mobile platform, a telescoping motor, two guide rails, and two linear guides. The linear guides are integrated with the mobile platform, whose telescoping movement is achieved by pulling two opposing cables via the telescoping motor. The cables are connected to the cable reel of the motor via multiple guide pulleys. The grabbing unit primarily consists of a grabbing motor, two grippers, and four cables to operate the grippers. It is capable of grabbing two distinct types of pipes (drill pipe and core pipe) with different diameters. The two grippers are vertically arranged to enhance grabbing reliability, with each gripper equipped with two clips. One end of each cable is connected to a gripper, and the other end is connected to the cable reel. The grabbing motor drives the cable reel to tighten the cables, causing the clips to close simultaneously and securely lock the pipe for manipulation. When the motor is reversed, the cables are loosened to release the gripper. Two torsion springs mounted in the grippers facilitate their retraction to the initial position, allowing the pipe to be handled by alternative mechanisms.

3.3. Coring component

The size of the coring component plays a crucial role in determining the manufacturing dimensions of the MWDR, and it stands as a significant design feature that directly influences the deep drilling capabilities of the proposed robotic sampling system. It primarily consists of a comprehensive set of drill tools, including specialized drill bits, drill pipes and core pipes, which work in tandem to facilitate efficient drilling operations. Additionally, the system is equipped with a wireline catcher designed to lift the core pipe from the drill pipe, ensuring smooth and controlled retrieval of the regolith samples.

3.3.1. Drill tool

The primary function of the wireline coring is a crucial factor in the drill tool design. This is a specialized tool used in conjunction with double pipes and a wireline. It is also a

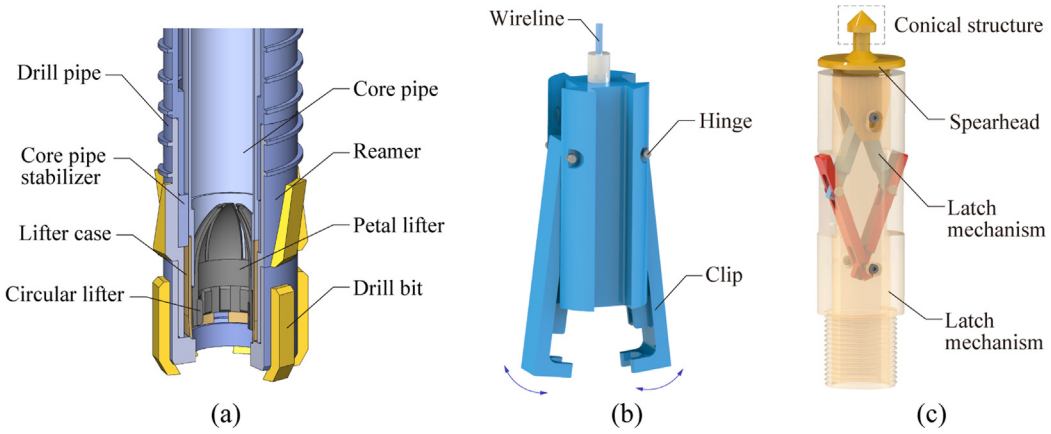


Fig. 6. Drill tools design. (a) Drill bit and core catcher. (b) Overshot. (c) Head assembly of core pipe.

critical component in extending the drill string. The drill tool (Fig. 6) primarily consists of the drill bit, drill pipe, overshot, and core pipe. The drill bit, primarily consisting of a bit body and a reamer, is mounted at the lower end of the first drill pipe. The drill bit is a hollow cylinder equipped with a set of blade teeth, designed to effectively cut through the regolith layer and expand the drilled borehole. The regolith inside the cutting annulus is pushed into the inner core pipe through the central cavity of the drill bit. This process can compress the drilled cuttings against the borehole wall, thereby increasing the diameter of the borehole during drilling. The connection between the bit body and the drill pipe is established through threads, which are also used to connect the reamer.

The drill pipe is a hollow cylindrical pipe featuring continuous spiral grooves on its outer surface, designed to facilitate cuttings removal during drilling. The lower end is fitted with external threads for connecting a drill bit or another drill pipe. An inner circular groove is positioned at the upper end of the pipe, designed to interface with the latch mechanism of the core pipe, thereby preventing unintended detachment during regolith extrusion. At the uppermost portion of the drill pipe, inner threads are present, allowing the assembly of multiple drill pipes to extend the overall length of the drill string. The overshot is mounted through a wireline to the wireline catcher which has a pair of deployable clips to lock the core pipe, with a return spring positioned in the middle. The core pipe consists of a head assembly for locking and unlocking the overshot, a hollow pipe for retaining core samples, and a core catcher for capturing intact regolith samples. When the core pipe is inserted into the drill pipe, a latch mechanism, including a four-bar linkage, extends due to gravity and secures the grooves of the drill pipe, effectively locking the core pipe in place during penetration. During the retrieval of the core pipe, the head assembly is raised by the overshot causing the spearhead to move upward by a certain distance. Subsequently, the latch mechanism folds inward, disengaging from the grooves and releasing the core pipe, allowing it to be extracted upward. The core catcher consists of a core pipe stabilizer, a lifter case, a petal lifter designed for loose-soft strata, and a circular lifter intended for hard rocks. The core pipe stabilizer is mounted on the reamer to ensure that the core pipe remains parallel to the drill pipe. The lifter case connects to the core pipe and houses the petal lifter within a tapered socket, thereby controlling the movement of the petal lifter. During core retrieval, as the drill string is raised from the bottom of the drilled borehole, the core exits the core pipe, compresses the petal lifter, and tightens it. Simultaneously, the circular lifter moves downward into the tapered socket within the petal lifter, allowing the grip marks on the inner surface of the circular lifter to engage the moving core.

3.3.2. Wireline catcher

The wireline catcher (Fig. 7) is used to move the overshot to lock and unlock the head assembly during core pipe retrieval. It consists of a deflection unit that aligns the overshot with the shaft of the drill pipe and an extraction unit that lifts and lowers the overshot within the drill string. The deflection unit mainly consists of a sliding platform, a winding wheel, a deflection motor, a base, two sliders, and two guide rails. The sliding platform is mounted on the two sliders and moved by a pair of antagonistic cables, which are wound around the output shaft of the deflection motor and pass through two fixed pulleys to adjust the horizontal position of the overshot. When the motor drives the winding wheel, one cable tightens while the other loosens, causing the sliding platform to move in one direction. The extraction unit mainly consists of an overshot, a wireline, a winch, a pulley, a limiter, and an extraction motor, functioning as a hoist. One end of the wireline is wound around the winch, while the other end is connected to the overshot through the pulley and limiter. During the core pipe retrieval operation, the extraction motor rotates to release the wireline and move the overshot downward toward the core pipe. When the overshot contacts the spearhead, the clips are opened by the conical structure atop it. Once the clips pass through the conical structure, they close due to the spring, locking the spearhead in place

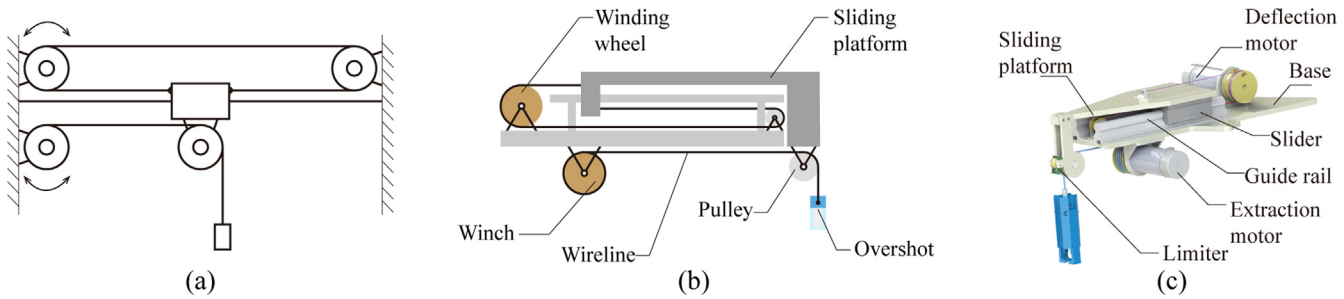


Fig. 7. Wireline catcher design. (a) Mechanism configuration. (b) Mechanism principle. (c) 3D model.

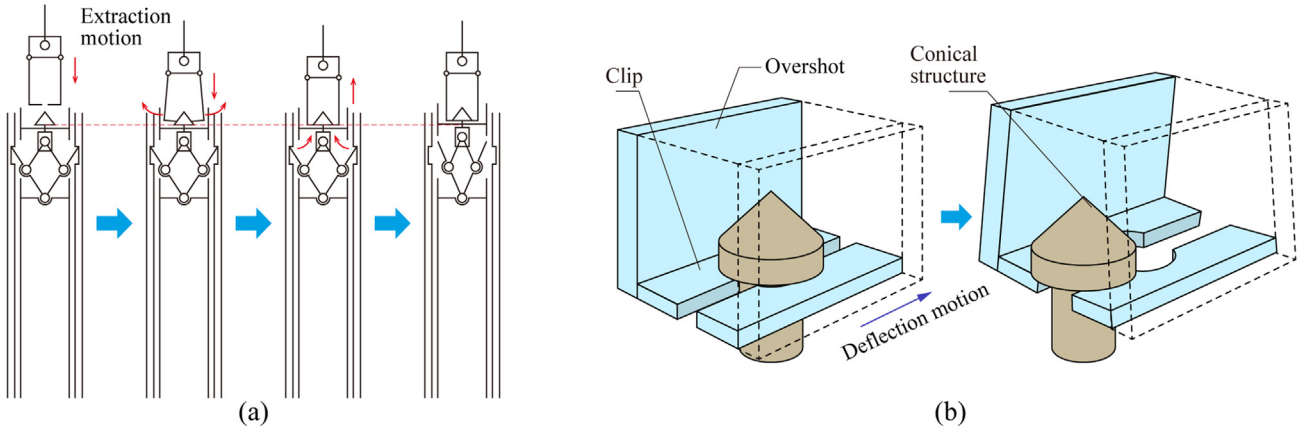


Fig. 8. Working principle of wireline catcher. (a) Extraction motion. (b) Deflection motion.

(Fig. 8a). This process of passing and locking is mainly influenced by the mass of the overshot, the elasticity of the spring, and the taper of the conical structure. The clip can be modeled as a lever with the hinge acting as the fulcrum, subjected to two opposing and equal forces. When the overshot secures the core pipe, the wireline is pulled to lift the overshot and core pipe out. After the core pipe is lifted out of the drill pipe, it is grabbed by the manipulator, and the sliding platform returns to its initial position, unlocking the connection between the overshot and core pipes (Fig. 8b).

3.4. Control system

The hardware control system (Fig. 9) of the MWDR comprises a console for human–computer interaction, a multi-axis motion controller (ZMC430N) for motion coordination, ten Elmo drives, and ten DC brushless motors. The console is capable of transmitting commands to the robot and receiving sensor feedback, including drilling parameters and status updates. The computer establishes communication with the multi-axis controller via Ethernet. The multi-axis controller functions as the primary station, with the ten motor drives serving as subordinate stations. EtherCAT is used as the serial communication protocol to facilitate multi-axis motion coordination between the master and slave stations. The multi-axis controller can transmit various commands, including motion mode, speed, acceleration, and position, to individual drives. It can also retrieve motor parameters, such as current, speed, acceleration, and position, from the drives. The motors are equipped with individual incremental encoders, which provide position and speed feedback. The EC-45 encoder has a count of 2048 pulses per turn (CPT), whereas the EC-60 flat encoder has a CPT of 4096. Ten DC brushless motors (Maxon) were employed in conjunction with Elmo drives (GMOLWHI 20/100EE) to control the motions of the 10-DOF MWDR. EC-60 motors, paired with GP-52C gearboxes having reduction ratios of 113, 26, 81, and 81, were used to drive the penetration, rotation, top clamping, and grabbing units, respectively. EC-45 motors, paired with GP-42C gearboxes (reduction ratio of 26), powered the bottom clamping, deflection, and extraction units. EC-45 motors, paired with GP-42C gearboxes (reduction ratio of 36), powered the telescoping and transferring units. An EC-max-40 motor, paired with a GP-52C gearbox (reduction ratio of 936), powered the rotary magazine.

4. Motion strategy

An MWDR must sequentially complete the complex motion steps based on the requirements of automatic regolith sampling operations. Therefore, coordinating multiple mechanisms to establish a well-structured sampling strategy is essential. For instance, during the sampling process, drill pipes must be sequentially connected for penetration

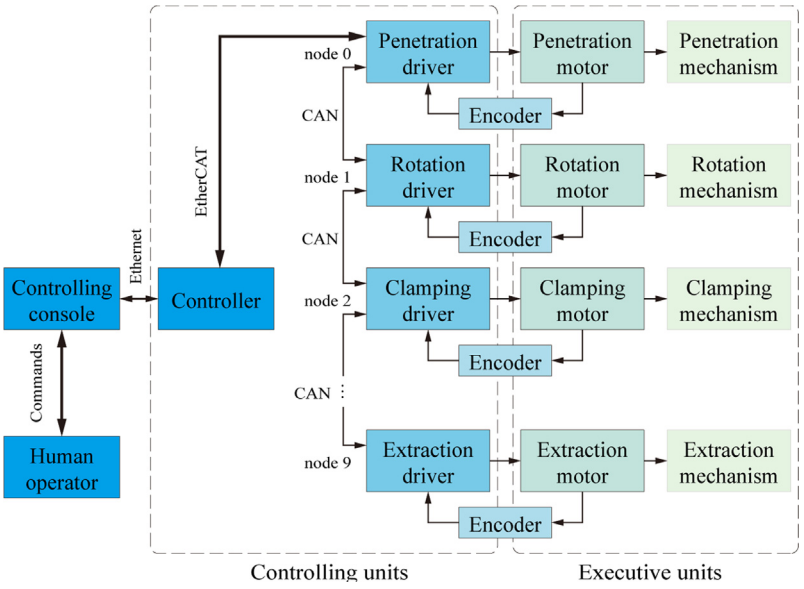


Fig. 9. Control system architecture..

and subsequently disconnected after sampling. Achieving this requires seamless coordination among multiple mechanisms, including the manipulator, drilling component, and rotary magazine. When examining the motion loop of a single drill pipe, a complete sequence is characterized by departure from and subsequent return to the rotary magazine. Meanwhile, a fundamental motion sequence for coring is integrated. Motion sequence planning involves designing a series of operational steps, such as clamping, transferring, rotating, and penetrating drill tools, to effectively execute the entire drilling and sampling process.

4.1. Mechanism state definition

The operation of the MWDR involves not only the straightforward transportation of drill pipes and core pipes between the rotary magazine and the drilling component for penetration, followed by their retrieval to the rotary magazine after sampling, but also seamless coordination with the wireline catcher for core pipe operations and with the drilling component for the connection and disconnection of drill pipes. If each individual motion step in the entire sampling process, from pipe selection to retrieval, were to be listed independently, it would involve dozens of sequences. To describe the complex sampling strategy, based on the DOF definitions outlined in Table 1, all mechanism states during the sampling process have been thoroughly analyzed and summarized (Table 2). Some mechanisms exhibit only two states: the initial state S_0 and the working state S_1, as observed in the clamping and releasing actions of the manipulator. Conversely, certain mechanisms exhibit three states: the initial state S_0, the working state S_1, and the working state S_2, as observed in the stopping, positive rotation, and negative rotation states of the rotation unit.

4.2. Motion sequence decomposition

To carry out a comprehensive sampling mission and elucidate the inherent coordination among mechanisms, the complete sampling process is broken down into seven sets of motion sequences. As shown in Fig. 10, these seven sets of motion sequences are executed based on a defined drilling and sampling strategy. As shown in Table 3, each set of motion sequences can be directly derived from the motion steps of individual mechanisms. The sequencing

Table 2
Mechanism state definition.

DOF	Symbol	Initial state S_0	Working state S_1	Working state S_2
Penetration	M_p	Top position	Bottom position	Middle position
Rotation	M_r	Stopping state	Positive rotation	Negative rotation
Clamping top	M_{ct}	Release	Clamping	None
Clamping Bottom	M_{cb}	Release	Clamping	None
Grabbing	M_g	Release	Grabbing	None
telescoping	M_{te}	Withdrawal	Extension	None
Transferring	M_{tr}	For pipe clamping	For drill pipe	None
Indexing	M_i	For drill pipe i (S_{0i})	For core pipe j (S_{1j})	None
Deflection	M_d	Withdrawal	Extension	None
Extraction	M_e	Withdrawal	Extension	None

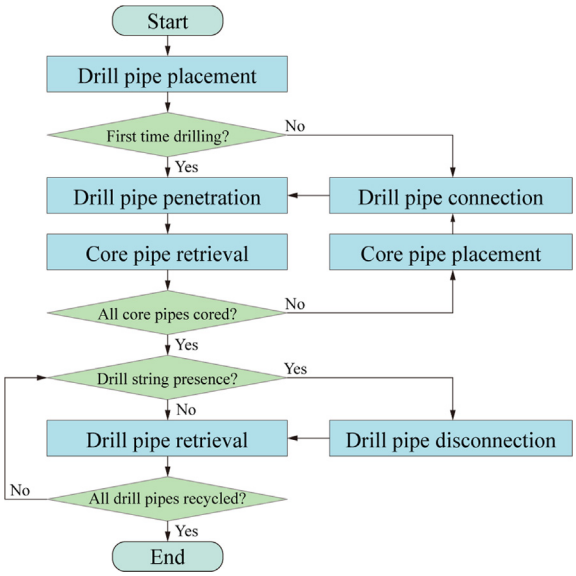


Fig. 10. Automatic drilling and sampling strategy..

of motion steps is meticulously designed to achieve optimal coordination among mechanisms, ensuring high efficiency. If a sequence of actions is not successfully executed, it may disrupt the standard planning process and potentially lead to the failure of the sampling mission. However, the current situation can be improved by incorporating specific sequences in between. For instance, lifting the core pipe from the drill pipe is prohibited before locking the overshot with the core pipe. In such cases, the overshot may be re-lowered to secure a lock with the core pipe. If locking the drill string remains unattainable, the entire drill string must be elevated to recover the core pipe, similar to the lifting coring method. Before initiating the sampling process, it is imperative that all mechanisms be reset to their initial positions. The rotary magazine stores empty drill pipes and core pipes. The initial drill pipe, along with a drill bit and a core pipe, is assembled and positioned at the grabbing location for operation. The initial position is shown in the first subfigure of the drill pipe placement sequence, excluding the indexing of the rotary magazine.

(1) *Drill pipe placement:* Before initiating each drilling operation, it is essential to clamp a fresh drill pipe to the drilling component. All drill pipes are stored within the rotary magazine. The indexing unit rotates the magazine to the designated grabbing position, while the manipulator transports the initial drill pipe to the drilling component for installation. The grabbing unit of the manipulator is positioned to align with this location and extended to securely grasp the pipe. The clippers secure the drill pipe, while the telescoping unit retracts smoothly to its original position to extract the pipe. The procedure for storing the drill pipe is the reverse of the placement process. This sequence consists of 11 steps (Fig. 11).

(2) *Drill pipe penetration:* Rotary drilling plays a pivotal role in the process of collecting regolith samples from the planetary surface. During this process, the drilling component undergoes both rotational and translational movements to drive the drill pipe downwards, creating a borehole and displacing the core material into the core pipe. In cases with softer layers, it is often more efficient to apply downward force directly onto the regolith layer without rotational movement.

(3) *Core pipe retrieval:* After each drilling cycle, it is essential to extract the core pipe containing the regolith samples from the lowest drill pipe and securely store it in the rotary magazine. First, the deflection unit extends the overshot to precisely align with the drilling shaft. The extraction unit then releases the overshot, which is gradually lowered along the inner wall of the drill pipes until it makes contact with and securely attaches to the spearhead. Next, the extraction unit extracts the wireline, causing the overshot to ascend a specific distance, along with the spearhead. The latch mechanism is then carefully manipulated to disengage the core pipe. The extraction unit subsequently reels in the wireline to raise the core pipe from the drill string. Once the core pipe is securely grasped by the manipulator, the deflection unit disengages the overshot, and then the manipulator transfers the core pipe to the rotary magazine for safe storage. This sequence consists of 15 steps (Fig. 12).

(4) *Core pipe placement:* After retrieving the preceding core pipe containing the core samples, the subsequent procedure involves inserting a fresh, empty core pipe into the drill string. In this procedure, the manipulator retrieves an empty core pipe from the rotary magazine and inserts it into the drill string, thereby positioning it at the lowest part of the drill pipe. The assembly process is achieved solely by the force of gravity, eliminating the need for any intervention from the drilling components.

(5) *Drill pipe connection:* To achieve a greater drilling depth, it is essential to continuously attach new drill pipes to the uppermost section of the existing drill pipe. The manipulator retrieves a fresh drill pipe from the rotary magazine and subsequently attaches it to the drilling component. Once the new drill pipe is securely fastened by the top clamping unit, it is rotated in a positive direction and gradually threaded downward into the drill string. As the newly acquired drill pipe is carefully threaded into the existing lower pipe, it is fully prepared for the subsequent phase of drilling operations.

(6) *Drill pipe disconnection:* All drill pipes must be retrieved and stored in sequence following the sampling process. When retrieving a drill pipe, it is essential to disconnect it from the drill string to facilitate handling. This process requires the use of the bottom clamping unit to securely hold the lower drill pipe in

Table 3
Motion sequence definition.

Function	Motion sequence
Drill pipe placement	$M_i \rightarrow S_{01}, M_{tr} \rightarrow S_1, M_{te} \rightarrow S_1, M_g \rightarrow S_1, M_{te} \rightarrow S_0, M_{tr} \rightarrow S_0,$ $M_{te} \rightarrow S_1, M_p \rightarrow S_2, M_{ct} \rightarrow S_1, M_g \rightarrow S_0, M_{te} \rightarrow S_0$
Drill pipe penetration	$M_p \rightarrow S_1$ and $M_r \rightarrow S_1, M_r \rightarrow S_0$
Core pipe retrieval	$M_{ct} \rightarrow S_0, M_p \rightarrow S_0, M_d \rightarrow S_1, M_e \rightarrow S_1, M_e \rightarrow S_0, M_{te} \rightarrow S_1, M_g \rightarrow S_1, M_{te} \rightarrow S_0, M_{tr} \rightarrow S_2,$ $M_i \rightarrow S_{11}, M_{te} \rightarrow S_1, M_g \rightarrow S_0, M_{te} \rightarrow S_0, M_{tr} \rightarrow S_0, M_d \rightarrow S_0$
Core pipe placement	$M_i \rightarrow S_{12}, M_{tr} \rightarrow S_2, M_{te} \rightarrow S_1, M_g \rightarrow S_1, M_{te} \rightarrow S_0, M_{tr} \rightarrow S_0, M_{te} \rightarrow S_1, M_g \rightarrow S_0, M_{te} \rightarrow S_0$
Drill pipe connection	$M_{cb} \rightarrow S_1, M_p \rightarrow S_2$ and $M_r \rightarrow S_1, M_r \rightarrow S_0, M_{cb} \rightarrow S_0$
Drill pipe disconnect	$M_p \rightarrow S_1, M_{ct} \rightarrow S_1, M_p \rightarrow S_2, M_{cb} \rightarrow S_1, M_p \rightarrow S_2, M_r \rightarrow S_2, M_r \rightarrow S_0$
Drill pipe retrieval	$(M_p \rightarrow S_1, M_{ct} \rightarrow S_1, M_p \rightarrow S_2)$, $M_{te} \rightarrow S_1, M_g \rightarrow S_1, M_{ct} \rightarrow S_0, M_p \rightarrow S_0, M_{te} \rightarrow S_0, M_{tr} \rightarrow S_1, M_i$ $\rightarrow S_{03}, M_{te} \rightarrow S_1, M_g \rightarrow S_0, M_{te} \rightarrow S_0, M_{tr} \rightarrow S_0$

position. The drilling component raises the drill string to the operational position, allowing the target drill pipe to be positioned just above the bottom clamping unit. The drilling component rotates the clamped drill pipe counterclockwise to disengage the threaded connection between the pipes. Secure clamping of the drill string is critical to prevent inadvertent sliding of the string into the borehole.

(7) *Drill pipe retrieval:* After completing the sampling mission, regolith samples are extracted from the core pipes, while the drill pipes remain in the borehole. The drill pipes can then be lifted to the surface by the drilling component and sequentially stored by the manipulator in the rotary magazine, similar to the core pipe retrieval process.

To more clearly illustrate the task allocation and synchronization logic of each mechanism at different sequence stages, the motion sequences are depicted as a logical flowchart (Fig. 13). In this flowchart, each functional unit is represented by an action node. The sequential and event-trigger relationships between nodes reflect both the logical and temporal flow of the entire process, from the placement of the drill pipe, through the drilling process, to the retrieval of the drill pipe.

5. Prototype development and experiments

To preliminarily verify the performance of the MWDR, the design has been implemented in a prototype system. The construction of the prototype system involves integrat-

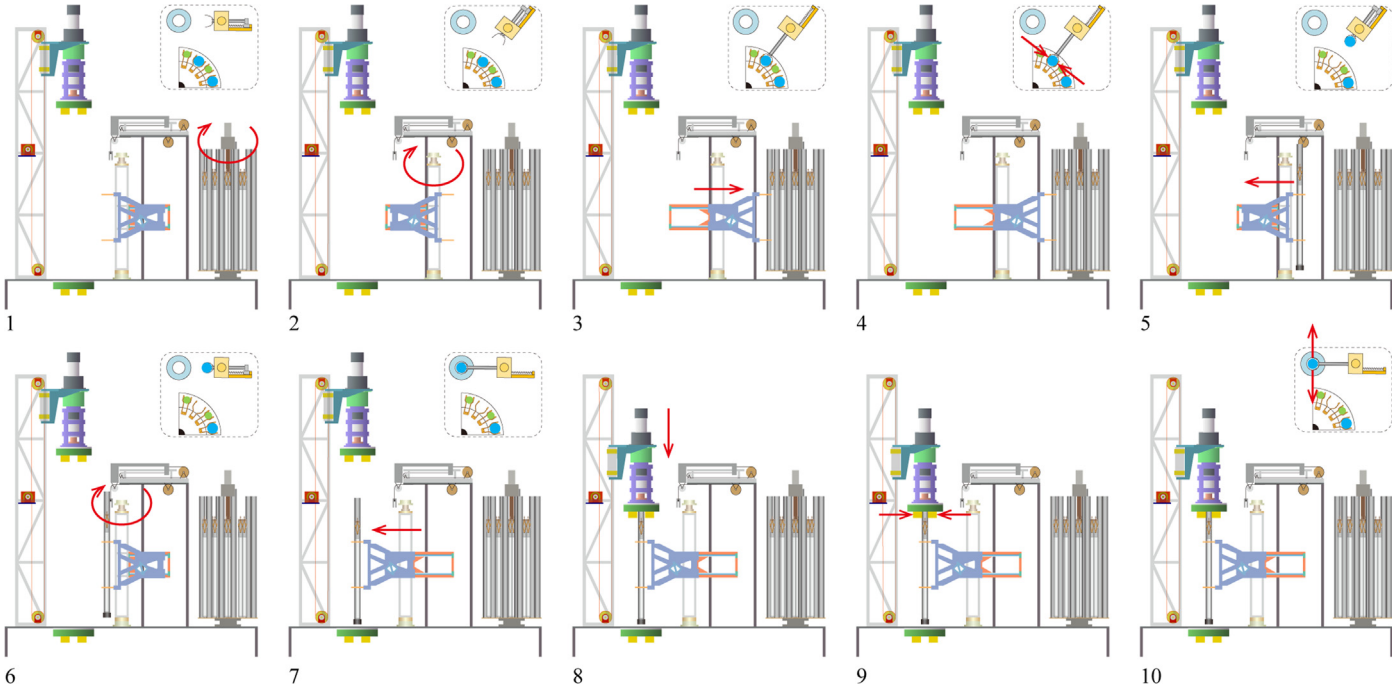


Fig. 11. Drill pipe placement sequence. The magazine rotates a drill pipe to the grabbing position (1). The manipulator rotates to align with the magazine (2), extends to approach the drill pipe (3), clamps the drill pipe (4), retracts to the initial position (5), rotates to align with the drilling component (6), and extends the drill pipe to the drilling shaft (7). The drilling component moves downward to approach the drill pipe (8) and clamps the drill pipe (9). The manipulator loosens the mounted drill pipe (10), and retracts to the initial position (11).

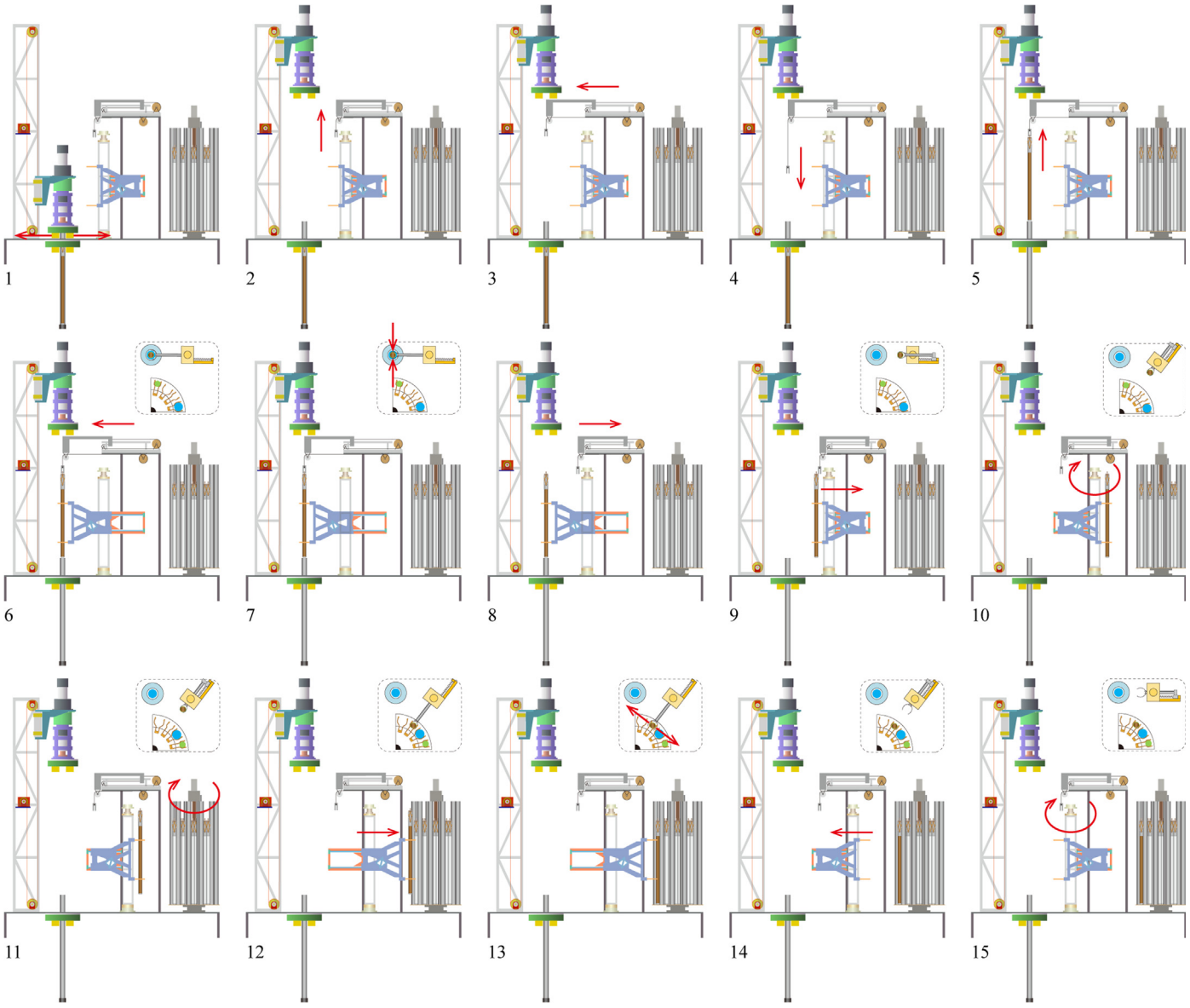


Fig. 12. Core pipe retrieval sequence. The drilling component loosens the penetrated drill pipe (1) and translates upward to its top position along the linear stroke (2). The deflection unit extends to align the overshot with the borehole (3) loosens the wireline to lower the overshot (4) and tightens the wireline to lift the overshot (5). The manipulator extends to the drilling component to approach the core pipe (6) and grabs the pipe (7). The deflection unit retracts to the initial position to separate the overshot from the core pipe (8). The manipulator retracts to the initial position with the core pipe (9) and rotates to align with the rotary magazine (10). The magazine rotates to index an empty space for the core pipe (11). The manipulator extends to the rotary magazine to insert (12) and release (13) the core pipe. The telescoping unit retracts to its initial position (14). The manipulator rotates to its initial position (15).

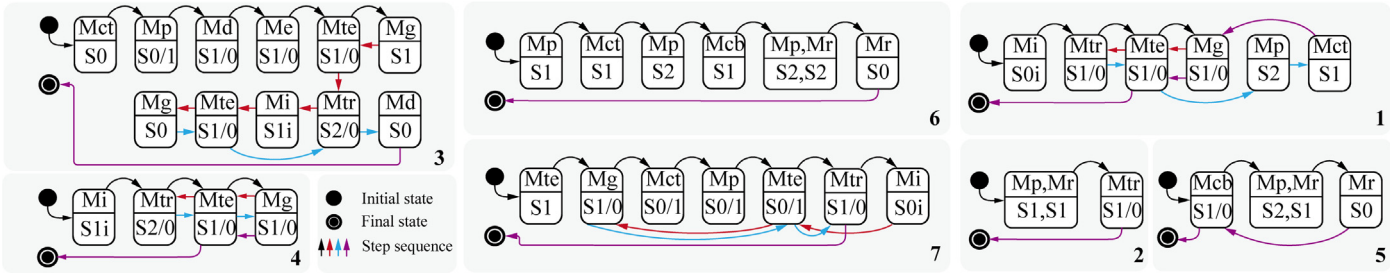


Fig. 13. Motion sequence logic flow. (1) Drill pipe placement, (2) Drill pipe penetration, (3) Core pipe retrieval, (4) Core pipe placement, (5) Drill pipe connection, (6) Drill pipe disconnection, and (7) Drill pipe retrieval. S[A/B] means the state alternates between S[A] and S[B], where S[A] happens first, and S[B] happens next time. S[A], and S[B] indicate states that happen simultaneously.

Table 4
Design parameters of the prototype.

Parameter	Value	Parameter	Value
Outer diameter of drill pipe (mm)	54	Transferring speed n_{tr} (rpm)	0∼120
Inner diameter of drill pipe (mm)	45	Indexing speed n_i (rpm)	0∼30
Outer diameter of core pipe (mm)	36	Penetration stroke S_p (mm)	830
Inner diameter of core pipe (mm)	30	Maximum WOB (N)	3000
Outer diameter of the reamer (mm)	62	Penetration speed v_p (mm/min)	0∼1400
Outer diameter of the bit body (mm)	54	Rotation speed n (rpm)	0∼120
Drill pipe thread length (mm)	30	Rotation torque (N·m)	0∼13
Clamping diameter d_c (mm)	36∼65	Deflection stroke S_d (mm)	150
Telescoping speed v_{te} (mm/min)	0∼1200	Deflection speed (mm/min)	0∼2400
Telescoping stroke S_{te} (mm)	0∼200	Extraction speed v_e (mm/min)	0∼2400

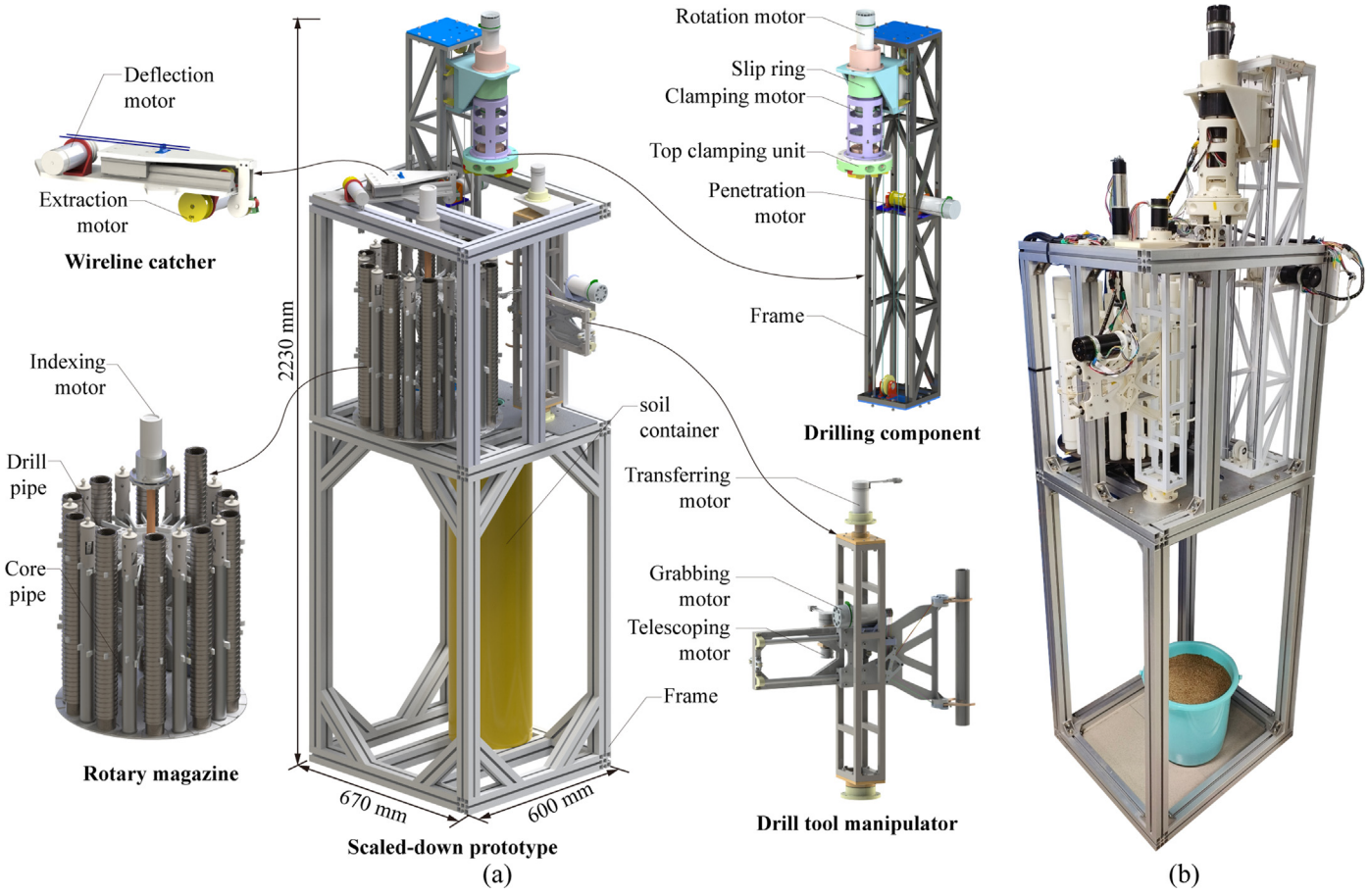


Fig. 14. Prototype development. (a) 3D model of the MWDR. (b) Prototype developed for ground tests.

ing and optimizing multiple key components, as well as the integration and debugging of the control system, to ensure a comprehensive evaluation of the feasibility and stability of the design. In the prototype system, the motion strategy of the multi-string and wireline coring is validated by a flexible mechanism layout and a reliable control system.

5.1. Prototype of MWDR

A prototype featuring a 0.5-m drill pipe was developed for experimental validation, based on the proposed mechanism design (Table 4 and Fig. 14). The primary objective of this research is to validate the design rationale and the motion characteristics of the mechanical system. The prototype has been fabricated and subjected to land-based testing. Most components were fabricated using 3D printing technology, utilizing ABS-like resin known for its excellent durability and resilience. Some critical components, which must endure substantial loads, underwent CNC machining. The test frame was constructed from aluminum profiles.

In accordance with the motion strategy proposed in this paper, the control system heavily relies on pre-planned static task sequences to coordinate the movements of various

mechanisms. These sequences are executed in an orderly and reliable manner through offline trajectory planning and a finite state machine model. Specifically, as previously mentioned, functions such as drill pipe placement, drill pipe drilling, and drill pipe retrieval are organized into seven primary action sequences. The motion paths and execution parameters for each action are pre-planned during the design phase and stored in the task buffer of the multi-axis controller. To prevent potential spatial or functional interference, these task sequences are prioritized and sequenced during the mechanism design and offline planning phases, ensuring the simultaneous execution of only non-conflicting actions. During the execution phase, when the control console sends task commands to the multi-axis motion controller, the controller determines the current sequence and, based on real-time feedback from sensors and encoders, periodically calculates target position, speed, and torque commands for each motor. These commands are transmitted to the corresponding drivers via the EtherCAT network within the same cycle, ensuring time synchronization between the actuators. If the current action fails to reach predefined position or torque thresholds, for example, during the sequence of penetration, when the clamping unit encounters resistance and fails to properly clamp the drill pipe, the control system will prevent subsequent rotational operations and classify the action as an "exceptional" state. In response, the system will trigger the reverse execution of the completed steps, retracting mechanical components to a safe position to prevent hardware damage or irreversible operational errors. As the system has not yet integrated global real-time path adjustment, it is classified as static scheduling logic. However, this approach has demonstrated high reliability and repeatability in standard multi-string drilling operations, as confirmed by experiments. This simple yet robust logic offers an effective means for verifying the design principles and motion characteristics of the drilling robot, while establishing a solid foundation for future exploration of dynamic task planning.

5.2. Experiments and results

Experimental tests were conducted in a laboratory environment using the developed robotic drill prototype to validate the functionality of the mechanical components. The motion of each mechanism was verified individually. For complex motions requiring the coordination of multiple mechanisms, pre-planned automatic drilling and sampling strategies were executed to validate the coordination of motion between the components. The operator sends commands to the robot via the control console, and the MADR executes the drilling experiment based on the predefined control program. The operational sequence precisely aligns with the drilling and sampling strategy. However, it is necessary to further elaborate on the operational details here.

The drill pipe placement motion was achieved through a coordinated effort involving the manipulator, the magazine, and the drilling component (Fig. 15a). Initially, the magazine indexed the first drill pipe to a position within reach of the manipulator. Subsequently, the transferring unit was activated to rotate and align with the drill pipe, followed by the extension of the telescoping unit until contact with the drill pipe was made, allowing for secure grasping. The manipulator then retracted to its initial position, drawing the drill pipe out and effectively disengaging it from the magazine. Next, the manipulator extended the telescoping unit and transported the drill pipe beneath the drilling component to align it with the component. As the drilling component descended, the clamping unit firmly secured the top of the drill pipe, preventing slippage. Finally, the manipulator was disengaged and returned to its initial position, without disrupting the ongoing drilling motion (Fig. 15b). It was observed that the movement states of the three key mechanisms could be easily adjusted, facilitating the sequential placement of drill pipes to achieve the desired drilling strategy using multiple pipes. Through meticulous parameterization of the motion program based on mechanism analysis and multiple pretests, the motion of each mechanism could be accurately predicted and expedited to transition to the next sequence, minimizing auxiliary time required for drilling.

After the drill pipe was installed, the drill pipe penetration motion was initiated, causing the drill pipe to descend into the soil and forcing samples into the core pipe for sealing (Fig. 15c). The clamping motor continuously applies torque in torque mode to the drilling component, effectively securing and rotating the drill pipe for penetration. In the initial experiments, a container filled with coarse sand was used to simulate planetary regolith, which is suitable for sampling. The container was placed below the drilling shaft, making direct contact with the descending drill pipe. During penetration, no slippage occurred between the drill pipe and the clamping unit, which confirmed its ability to securely hold the drill pipe and ensure successful penetration and rotation. The drill pipe penetrated 100 mm into the sand as a result of the combined penetration and rotation movements of the drilling component. During this process, sand cuttings were transported upward along the annular gap of the cutting edge and collected within it. However, if these cuttings were not cleared, the WOB would significantly increase with drilling depth. Due to the limitations in the strength of the 3D-printed drill tools, the drilling test was terminated at this depth. For multi-pipe drilling and retrieval, the (dis) connection motion of multiple drill pipes was required (Fig. 15d). The bottom clamping unit initially secured the lower drill pipe before connecting a new one. Subsequently, the drilling component engaged with the new drill pipe, moving downward and rotating in the positive direction according to the thread rotation and length, thus effectively completing the drill pipe connection. It is important to note that the disconnection process for the drill pipe is the reverse of this step.

The first core pipe was placed inside the first drill pipe, and the latch mechanism locked the drill pipe in an

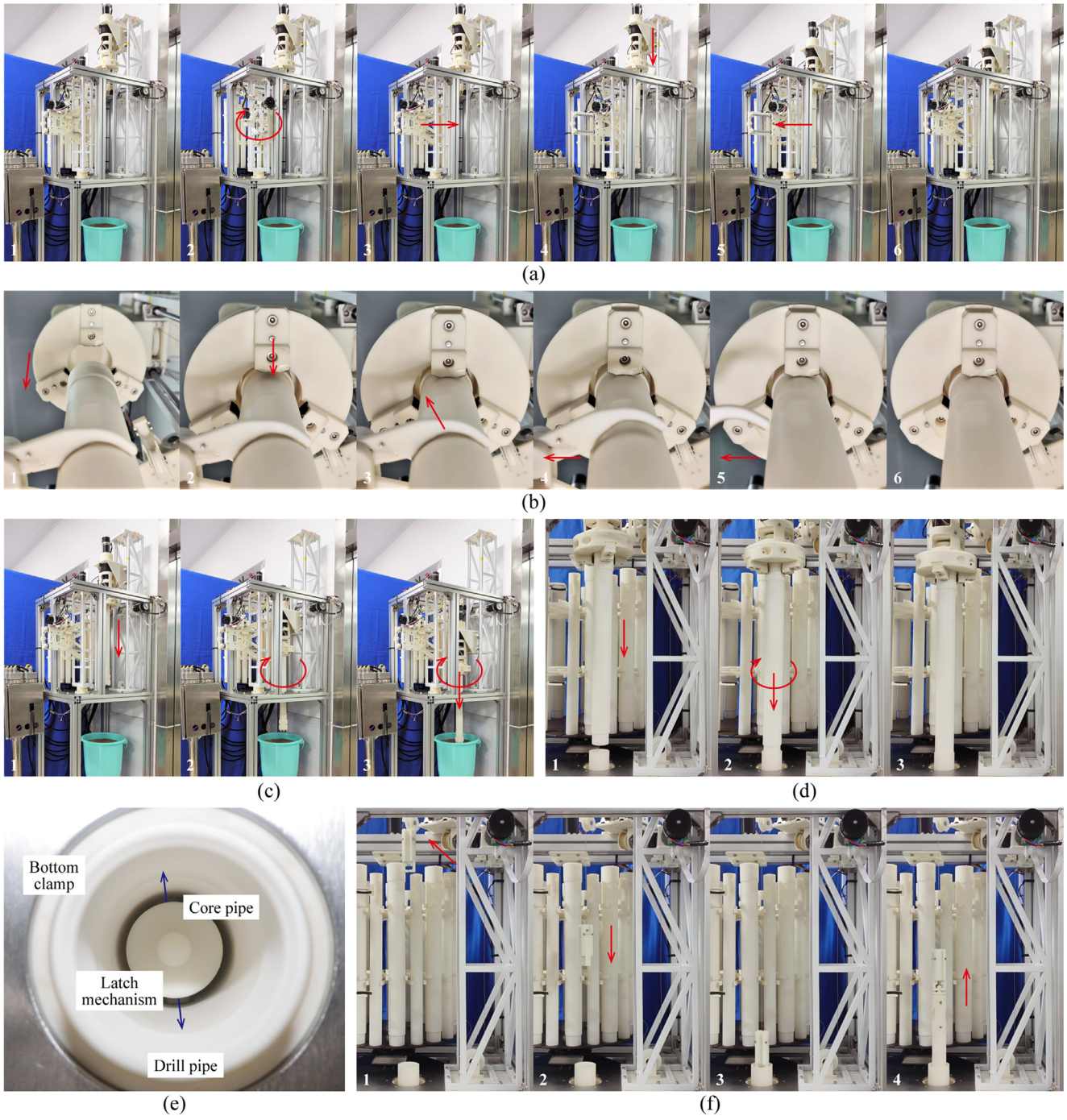


Fig. 15. Snapshots during multi-string automatic drilling. (a) Complete process of drill pipe placement. (b) Localized detail of drill pipe placement. (c) Complete process of drill pipe penetration. (d) Complete process of drill pipe connection. (e) Core pipe locks into the drill pipe. (f) Complete process of core pipe retrieval.

expanded state (Fig. 15e). When the drilling component completes its maximum drilling stroke, the motion for retrieving the core pipe can be initiated (Fig. 15f). The overshot was delivered by the deflection unit to align with the drilling shaft. The wireline was then released to lower the overshot rapidly until both clips locked onto the conical structure. The latch mechanism was retracted from its expanded state to release the core pipe from the drill pipe. The overshot was steadily lifted to the surface by the wireline catcher at the same speed as the downward motion. No accidental disengagement occurred throughout the process. Subsequently, the core pipe was placed back into the magazine for storage, following the motion sequence for core pipe retrieval.

In addition to verifying the multi-string automatic drilling process, a drilling test was conducted using a 3D-printed spiral drill tool in subsequent experiments to evaluate the drilling performance of the MWDR further. In

this test, the coarse sand in the original container was replaced with CUG-1A lunar regolith simulant to simulate planetary surface regolith. The WOB was limited to 200 N to prevent structural damage due to excessive load. Experimental data were collected via built-in motor sensors and converted to corresponding physical quantities using the aforementioned equations, which transformed the measured current, position, and velocity values. It should be noted that this experiment primarily aimed to verify the feasibility of the system working principle and does not represent the performance limits of the drilling system.

As shown in Fig. 16, a complete drilling experiment comprised three phases: drill bit approach to the regolith simulant, drilling procedure, and drill bit lifting. In this experiment, the penetration speed was set at 25 mm/min, and the rotation speed was maintained at 60 rpm. As shown in Fig. 17, after the drill pipe was installed, the penetration unit drove the drill bit to descend above the bottom clamping unit and slowly approached the surface of the regolith simulant. During this process, the WOB showed slight fluctuations due to friction from the rubber blocks in the bottom clamping unit. Once the drill bit contacted the surface, the drilling operation began according to the predetermined procedure. Under constant penetration conditions, the drill bit gradually penetrated the regolith simulant as drilling progressed. It was observed that the WOB and rotation torque exhibited an exponential growth trend, consistent with the mechanical characteristics observed in most previous lunar regolith drilling experiments. The regolith cuttings were transported upward through the cutting edges and spiral grooves of the drill tool, gradually forming a conical deposition block at the borehole surface. When the drill bit was fully embedded in the regolith simulant, the deposition block expanded outward from within the borehole as the cuttings accumulated, eventually accumulating at the top. When the maximum WOB reached the preset safety threshold of 200 N, the drilling operation ceased. When the penetration motor

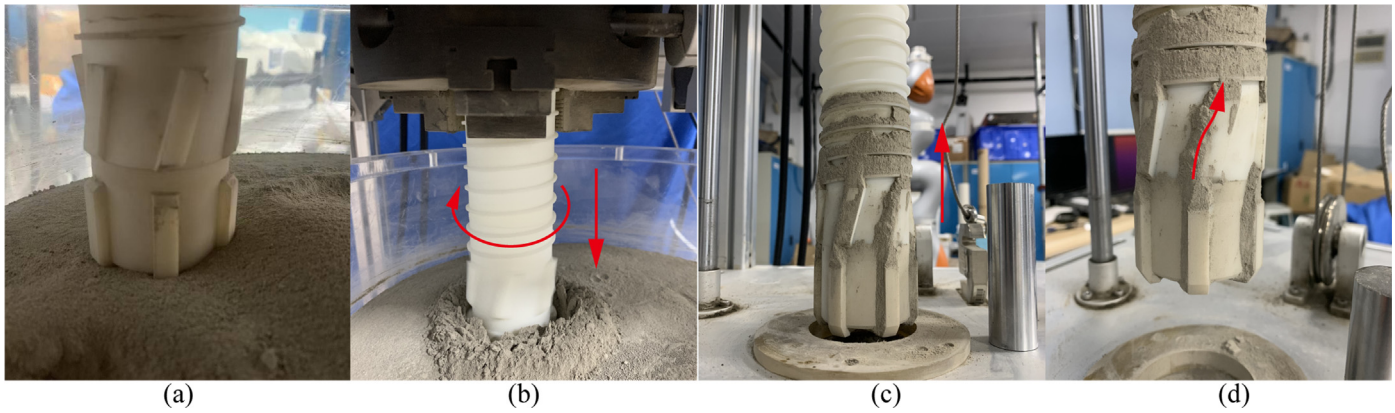


Fig. 16. Snapshots during drilling. (a) drill bit approach to the regolith simulant. (b) Drilling procedure. (c) Drill bit lifting. (d) Cuttings flow on the drill tool.

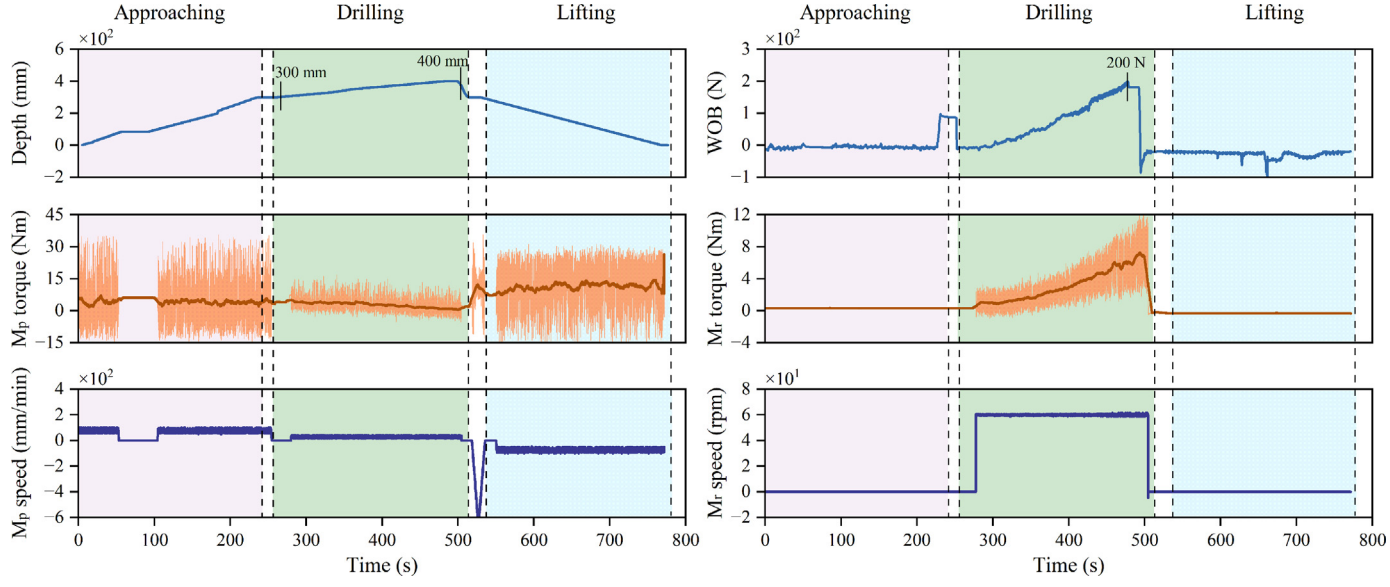


Fig. 17. Representative drilling result curves for a penetration speed of 25 mm/min and a rotation speed of 60 rpm.

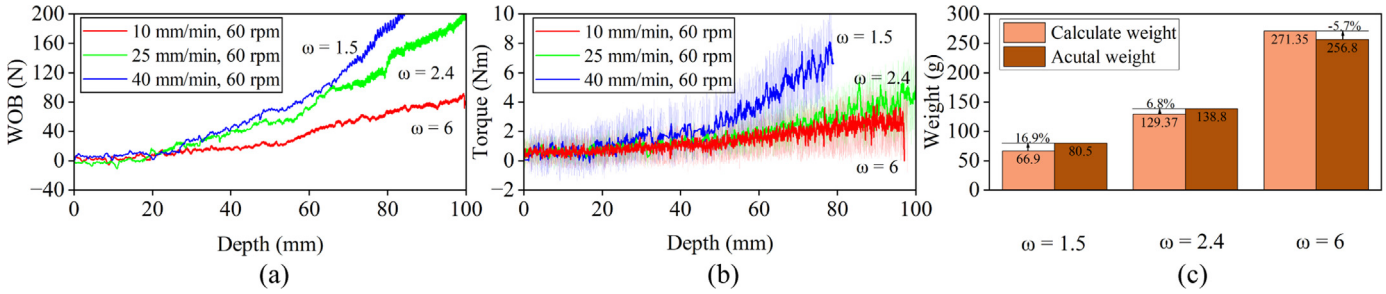


Fig. 18. Drilling performance analysis. (a) WOB curves for different drilling ratios. (b) Rotation torque curves for different drilling ratios (10-pt average). (c) Sample weight comparison between indirectly calculated using sensing parameters and actual weight.

stopped, the compacted regolith simulant and the tensioned steel cable generated a reverse force on the motor, causing the penetration unit to undergo reverse motion. Subsequently, the lifting process began at a constant speed. During lifting, friction between the borehole wall and the drill bit resulted in a reverse peak in drilling force, which gradually stabilized as the drill bit exited the regolith layer.

Furthermore, additional drilling performance tests were conducted under the same rotation speed but with different penetration speeds (v_p = 10, 25, 40 mm/min). As shown in Fig. 18 a and b, the drilling force (WOB and rotation torque) increased as the drilling ratio ω (rotation speed/penetration speed) increased at a constant drilling depth during the drilling process. This indicated that, under certain conditions, increasing the drilling ratio ω of the drill pipe helped reduce drilling resistance, thereby enabling deep drilling operations with lower loads. During penetration, a portion of the regolith samples was compressed into the core pipe. Upon experiment completion, the collected regolith samples in the core pipe were weighed and compared with the theoretical calculations. Due to manual handling, some regolith samples were accidentally lost during the weighing process. As shown in Fig. 18c, the average difference between the measured regolith mass and the converted mass from sensor data was only 6%, which partially demonstrates the necessity of using motor sensors for physical measurements in the drilling system.

6. Conclusions and future work

An unmanned sampling of planetary regolith heavily relies on the automation of robotic sampling systems, which must perform multi-step and multi-sequence collaborative operations. To tackle potential challenges associated with deep sampling on extraterrestrial surfaces, this paper proposes an MWDR, integrating wireline coring technology into the multi-string drilling robot for planetary regolith sampling for the first time. The MWDR aims to obtain bottom samples during the multi-string drilling process quickly, segment planetary regolith into core samples for preservation, and recover the drill pipe upon reaching the desired drilling depth. To achieve this, the study presents a detailed design of the MWDR mechanism and control system, proposes a feasible motion strategy, and develops a prototype. The functionality of the system has been validated through multiple experiments, and its performance has been evaluated. However, the MWDR for planetary regolith sampling constitutes a complex system that demands consideration of multiple factors. Due to cost and testing constraints, this study concentrates on designing a viable mechanical system. Future work must address issues such as optimizing component materials to increase strength and rigidity for withstanding higher drilling loads and improving motion planning to enhance coordination between mechanisms to improve sampling efficiency.

Declaration of competing interest

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